

8. QUEUING THEORY

NBS

THEORETICAL QUESTIONS

1. 2070 Old Q.No. 5

What is a queue? Describe various characteristic of queue and queuing system. [10]

2. 2067 Q.No. 5

What is a queue? Describe various operation characteristics of a queue and queuing system. [10]

3. 2058 Q.No. 6

What is queue? Describe the various operation characteristics of queue and queuing system. [10]

NUMERICAL QUESTIONS

1. 2070 Q.No. 10a

① In the post office, customers arrive in the ticket window is 10 per hour. The service rate of the window is 15 customers per hour. Find the probability that (i) the employee in the counter is idle and (ii) customer arriving has to wait for the service. [5]

$$(i) \left(1 - \frac{\lambda}{\mu}\right) \quad (ii) \frac{\lambda}{\mu}$$

Ans: $\frac{1}{3}, \frac{2}{3}$

2. 2069 Q. No. 7b

② If the arrival of customers for service is 10 per hour and the average service rate is 12 customers per hour.

- What is the probability that there are 5 customers in the system?
- Find the length of the queue?

[5]

Ans: (i) 0.07 (ii) 4.17 customers

3. 2065 Q.No. 9 b

③ A person repairing radios finds that the time spent on the radio sets has an exponential distribution with mean 20 minutes. If the radios are required in the order in which they come in and their arrival is approximately Poisson with an average rate of 15 for 8-hour day, what is the repairman's expected idle time each day? How many jobs are ahead of the average set just brought in? [10]

Ans: 3 hrs; $5/3 \approx 2$ sets

4. 2064 Q.No. 8 b

④ On the average 96 patients per 24 hours day require the service of an emergency clinic. Also on the average, a patient requires 10 minutes of active attention. Assume that the facility can handle only one emergency at a time. If this situation satisfy all the conditions for apply queuing theory, find the average (expected) queue length and the waiting time for the patient to be served. [10]

5. 2063 Q.No. 6

⑤ What is maintenance management? How can theory of queuing be utilized in maintenance management system? Cars arrive in a store (single channel) at an average rate of six per hour and service rate for the channel is 10 per hour. What is the length of the queue? [10]

Ans: 0.90

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9. 2061 Q.No. 9 (b) (ii)

- 6 A departmental store has a single cashier. During rush hours, customers arrive at the rate of 20 customers per hour. The average number of customers that can be processed by the cashier is 24 per hour. Assume that the conditions for use of single-channel queuing model apply. What's the (a) probability that the cashier is idle? (b) average no. of customers in queuing system? (c) average time a cashier spends in the system? (d) average time a customer spends in the queue waiting for services?

Ans: (a) $P_0 = 0.17$ (b) 5 customers (c) 15 minutes (d) 12.5 minutes [6]